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Motivic Zeta-Function of a Severi-Brauer Variety

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Abstract

We study the motivic zeta-function $Z_X(t)$ of an algebraic variety X. The main result is that for smooth complete surfaces in characteristic zero rationality of the motivic zeta-function is a birational invariant. We also prove some theorems on motivic zeta-function of Severi-Brauer varieties and compute the motivic zeta-function of a Cartesian power of a smooth conic.

Contents

Introduction **3**

1 Preliminaries **4**
1.1 Grothendieck ring of algebraic varieties 4
1.2 Motivic zeta-function 5

2 Birational invariance of rationality of motivic zeta-function for surfaces **6**

3 Motivic zeta-function of Severi-Brauer varities **8**
3.1 Severi-Brauer varities 8
3.2 Motivic zeta-function 10

4 Motivic zeta-function of Cartesian power of a conic **12**

References **13**

Introduction

Motivic zeta-function $Z_X(t)$ is a fundamental object of the modern algebraic geometry and number theory. It is a natural generalization of another important object which is Hasse-Weil zeta-function of a variety X defined over a finite field $\mathbb{F}_q$:

$$Z_{HW}(X,t) = \exp\left(\sum_{m\geqslant 0}\frac{N_m}{m}t^m\right) \in \mathbb{Q}[[t]],$$

where $N_m := |X(\mathbb{F}_{q^m})|$ is the number of $\mathbb{F}_{q^m}$-points of X.
The motivic zeta-function is defined as follows:

$$Z_X(t) := 1 + \sum_{k=1}^{\infty}[\mathrm{Sym}^k(X)]t^k$$

for a quasiprojective algebraic variety X. The word *motivic* means that $Z_X(t)$ is defined with coefficients in the Grothendieck ring of algebraic varieties $K_0(Var_k)$ over k. Here we denote by $[X]$ the class of X in $K_0(Var_k)$.

One of the most celebrated results about the Hasse-Weil zeta-function is the proof of the Weil conjectures which describe a lot of properties of $Z_{HW}(X,t)$, including its rationality. Kapranov suggested that the motivic zeta-function $Z_X(t)$ should be a rational function too. It is indeed rational for a smooth geometrically connected projective curve [L, $Theorem10$] (it was firstly proved by Kapranov for curves containing a rational point point over the base field, and then Litt generalized the result), but in [LL] it was proved that already for a complex surface S rationality of the motivic zeta-function $Z_S(t)$ is equivalent to $\kappa(S) = -\infty$, where $\kappa(S)$ is the Kodaira dimension. It is natural to ask to ask to what extent these results can be generalized. Our work is devoted to studying the motivic zeta-function of some classes of algebraic varietes, mostly Severi-Brauer ones.
The text is organized as follows:

In section 1 we give some basic definitions and facts about the Grothendieck ring of algebraic varieties $K_0(Var_k)$ and the motivic zeta-function $Z_X(t)$.

In section 2 we prove the main theorem of the thesis:

Theorem. *Let X be a smooth complete algebraic surface over a field of* $\operatorname{char} k = 0$. *If $Z_X(t)$ is rational, then $Z_Y(t)$ is rational for any smooth complete algebraic surface Y which is birational to X.*

In section 3 we study the motivic zeta-function of Severi-Brauer varieties. The main result of this part is that $Z_X(t)$ is rational over $K_0(Var_k)/(\mathbb{L})$ for any Severi-Brauer variety X, where $\mathbb{L}$ is the class of affine line. We also prove some theorems on links between rationality of the motivic zeta-function of different Severi-Brauer varieties.

In section 4 we prove the following theorem:

Theorem. *Let C be a smooth conic. Then*

$$Z_{C^{n+1}}(t) = \prod_{k=0}^{n} Z_C^{\binom{n}{k}}(\mathbb{L}^k t)$$

and it is a rational function.

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1 Preliminaries

Convention: all algebraic varieties are defined over an arbitrary field k unless the contrary is stated.

1.1 Grothendieck ring of algebraic varieties

Definition 1.1. *The Grothendieck ring of algebraic varieties over k* denoted by $K_0(Var_k)$ as an abelian group is generated by the set of isomorphism classes of algebraic varieties with relations of the form

$$[X] = [Y] + [X \setminus Y]$$

where Y is a closed subvariety in X (we denote by $[X]$ class of the variety X in $K_0(Var_k)$). The multiplication (associative and commutative) in this ring is given by

$$[X] \cdot [Y] = [(X \times Y)_{\mathrm{red}}]$$

It is clear that $[\emptyset] = 0$ and $[\mathrm{Spec}(k)] = 1$ in $K_0(Var_k)$.

We denote by $\mathbb{L} \in K_0(Var_k)$ class of the affine line in the Grothendieck ring of varieties over k.

Example 1.1. $[\mathbb{A}^n] = [(\mathbb{A}^1)^n] = [\mathbb{A}^1]^n = \mathbb{L}^n$.

Example 1.2. $[\mathbb{P}^n] = 1 + \mathbb{L} + \mathbb{L}^2 + \cdots + \mathbb{L}^n$.
It follows from the decomposition $\mathbb{P}^n = \mathrm{Spec}(k) \sqcup \mathbb{A}^1 \sqcup \mathbb{A}^2 \sqcup \cdots \sqcup \mathbb{A}^n$ and induction on n because $\mathbb{P}^1 \setminus \mathbb{A}^1 = \mathrm{Spec}(k)$ and $\mathbb{P}^n \setminus \mathbb{A}^n \cong \mathbb{P}^{n-1}$ for all $n \geqslant 1$.

Proposition 1.1. *[M, Proposition 7.4] If $f\colon X \to Y$ is piecewise trivial with fiber F, i.e. there is a decomposition $Y = Y_1 \sqcup \cdots \sqcup Y_r$ with all Y_i are locally closed in Y such that $f^{-1}(Y_i) \cong Y_i \times F$, then $[X] = [Y] \cdot [F]$ in $K_0(Var_k)$.*

Corollary 1.1. *Let $E \to Y$ be a vector bundle of rank n on an algebraic variety Y, then*

$$[E] = [Y] \cdot \mathbb{L}^n \tag{1}$$
$$[\mathbb{P}(E)] = [Y][\mathbb{P}^{n-1}] = [Y](1 + \mathbb{L} + \cdots + \mathbb{L}^{n-1}) \tag{2}$$

Whilst the $K_0(Var_k)$ is still understood very poorly, it turns out that $K_0(Var_k)/(\mathbb{L})$ have a quiet nice description in terms of stable birationality. Recall that two algebraic varieties are called *stably birational* if there exist $n, m \geqslant 0$ such that $X \times \mathbb{P}^n$ is birational to $Y \times \mathbb{P}^m$. We denote this by $X \overset{s}{\sim} Y$.

Theorem 1.1. *[M, Theorem 7.20] $K_0(Var_k)/(\mathbb{L})$ is generated by classes of stable birational equivalence of irreducible algebraic varieties when* $\mathrm{char}\, k = 0$.

In the original paper by M. Larsen and V.Lunts Theorem 1.1 was proved for algebraically closed fields. It turns out that each step of the proof still works over an arbitrary field of $\mathrm{char}\, k = 0$ because the assumption of the algebraical closure was used only to claim that a birational map between smooth projective varieties can be realized as a composition of blow-ups and blow-downs with smooth centres, but it is true for any field of $\mathrm{char}\, k = 0$ according to [AKMW, Theorem 0.3.1].

1.2 Motivic zeta-function

Definition 1.2. Let X be a quasiprojective algebraic variety, then the *motivic zeta-function* $Z_X(t) \in 1 + tK_0(Var_k)[[t]]$ is the formal power series

$$Z_X(t) := 1 + \sum_{k=1}^{\infty} [\mathrm{Sym}^k(X)]t^k \tag{3}$$

Even though the symmetric powers $\mathrm{Sym}^k(X)$ are well-defined only for quasiprojective varieties there is the following

Proposition 1.2. *[M, Proposition 7.28] The motivic zeta-function defines a homomorphism*

$$K_0(Var_k) \to 1 + tK_0(Var_k)[[t]]$$

that is, if $[X] = [Y] + [Z]$, *then* $Z_X(t) = Z_Y(t) \cdot Z_Z(t)$

Example 1.3.

- $\mathrm{Sym}^n(\mathrm{Spec}(k)) \cong \mathrm{Spec}((k^{\otimes n})^{S_n}) \cong \mathrm{Spec}(k)$ for all n so we have

$$Z_{\mathrm{Spec}(k)}(t) = 1 + \sum_{k=1}^{\infty} t^k = \frac{1}{1-t}$$

- $\mathrm{Sym}^n(\mathbb{A}^1) \cong \mathbb{A}^n$ for all n so we have

$$Z_{\mathbb{A}^1}(t) = 1 + \sum_{k=1}^{\infty} [\mathbb{A}^k]t^k = \frac{1}{1-\mathbb{L}t}$$

One of the main properties of the motivic zeta-function which allows computing it in many examples is the following:

Proposition 1.3. *Let X be an algebraic variety, then*

$$Z_{X\times\mathbb{A}^n}(t) = Z_X(\mathbb{L}^n t) \tag{4}$$

Moreover, for any projective bundle $E \to X$ of rank n one has

$$Z_E(t) = \prod_{k=0}^{n} Z_X(\mathbb{L}^k t) \tag{5}$$

Proof: The first statement is the [M, Proposition 7.32] while the second one follows directly from the first one, Example 1.2 and the fact that motivic zeta-function is a homomorphism. ■

Example 1.4. $Z_{\mathbb{A}^n}(t) = \dfrac{1}{1-\mathbb{L}^n t}$ and $Z_{\mathbb{P}^n}(t) = \dfrac{1}{(1-t)(1-\mathbb{L}t)\dots(1-\mathbb{L}^n t)}$ by taking $X = \mathrm{Spec}(k)$ in (4) and (5) respectively.

Remark. We mostly want to study rationality questions of motivic zeta-function. According to [M, Theorem 7.21] $K_0(Var_k)$ contains zero-divisors so one should be careful about the definition of a rational function in $K_0(Var_k)[[t]]$.

Definition 1.3. $f(t) \in K_0(Var_k)[[t]]$ is called *rational* if there exists a non-zero polynomial $g(t) \in K_0(Var_k)[t]$ such that $f(t)g(t) \in K_0(Var_k)[[t]]$ is a polynomial.

Proposition 1.4. *[LL, Lemma 3.5] If $f(t) \in 1 + tK_0(Var_k)[[t]]$ is rational, then $f^{-1}(t)$ is also rational.*

The following property is also a useful tool for proving rationality of motivic zeta-function:

Proposition 1.5. *[LL, Lemma 3.1] Let X be an algebraic variety and $Y \subset X$ be its closed subvariety, then if any of two $Z_X(t)$, $Z_Y(t)$ and $Z_{X\setminus Y}(t)$ are rational, then the third one is rational as well.*

Some examples of rational motivic zeta-functions are provided by computations in Examples 1.3 and 1.4. The next theorem provides an example of a class of algebraic varieties with a rational motivic zeta-function.

Theorem 1.2. *[L, Theorem 10] Motivic zeta-function $Z_C(t)$ of a smooth, projective and geometrically connected curve C is a rational function with invertible numerator and denominator.*

One more example of a class of algebraic varieties with rational motivic zeta-function will be provided in the section 4.

2 Birational invariance of rationality of motivic zeta-function for surfaces

This section is devoted to the proof of the main result of the thesis:

Theorem 2.1. *Let X be a smooth complete algebraic surface over a field of* $\operatorname{char} k = 0$. *If $Z_X(t)$ is rational, then $Z_Y(t)$ is rational for any smooth complete algebraic surface Y which is birational to X.*

The theorem shows that the rationality of the motivic zeta-function is a birational invariant for smooth projective algebraic surfaces in characteristic 0. Using this fact, one can check the rationality of the motivic zeta-function on a smooth minimal model of the surface. In [LL] it was proved that if the Kodaira dimension of the surface is non-negative, then the motivic zeta-function is irrational. It follows that the only possibility for X to have a rationval $Z_X(t)$ is Mori fiber spaces, which are del Pezzo surfaces or conic bundles. Rationality of motivic zeta-functions for del Pezzo surface of degree 8 of the form $C \times C$ is proved in Theorem 4.1, while degree 9 del Pezzo surfaces are precisely Severi-Braur surfaces, and we will prove some theorems on their motivic zeta-functions in the section 3.

Let us prove first two more lemmas that will be necessary in the proof of the Theorem 2.1:

Lemma 2.1. *Let X be a smooth algebraic variety and $Y \subset X$ its smooth subvariety. Then the motivic zeta-function $Z_E(t)$ of the exceptional divisor E of $Bl_Y X$ is rational if $Z_Y(t)$ is rational.*

Proof: Centre of the blow-up Y is smooth, hence it is a local complete intersection and then the exceptional divisor E is $\mathbb{P}^r$-bundle over Y for some r. It follows from (5) that

$$Z_E(t) = \prod_{k=0}^{r} Z_Y(\mathbb{L}^k t)$$

so if $Z_Y(t)$ is rational then $Z_E(t)$ is also rational. ■

Lemma 2.2. *Motivic zeta-function $Z_X(t)$ is rational for any zero-dimensional variety X over a perfect field k.*

Proof: It is enough to prove the theorem for each connected component of X because the motivic zeta-function is a homomorphism. They are of the form $\mathrm{Spec}(L)$ where $L \supset k$ is a finite extension of the base field k. The base field k is perfect, hence L is a separable extension and let $F \supset L$ be its Galois closure with the Galous group $G = Gal(F/k)$. Any tensor power $F^{\otimes n}$ of F is a finite product of finite separable extensions of k (i.e. a finite étale k-algebra) and hence corresponds (via the fundamental theorem of Galois theory) to a finite G-set. Using this fact, we translate the initial problem to the language of finite G-sets.

Let $B(G)$ be the *Burnside ring of* G which is an abelian group generated by isomorphism classes of finite G-sets, with addition given by disjoint union. Ring multiplication in this ring is given by Cartesian product of the underlying sets. As we explained above, each finite G-set corresponds to a finite étale k-algebra, i.e. there is a homomorphism $B(G) \to K_0(Var_k)$ sending an isomorphism class of finite G-set to the class of Spec of the corresponding finite étale k-algebra in $K_0(Var_k)$. It follows that it is enough to prove that

$$1 + \sum_{n=1}^{\infty} [\mathrm{Sym}^n(Y)] t^n \tag{6}$$

is rational over $B(G)$, where Y is a finite G-set and we denote by $[Y]$ its class in $B(G)$. Any finite G-set Y is a finite disjoint union of G-sets each of which is isomorphic to G/H_i for some subgroups $H_i \subset G$, and it is enough to prove the claim for G-sets of such form.

$\mathrm{Sym}^n(Y) \in B(G)$ is defined as a set of all multisubsets of Y of cardinality n, that is:

$$\mathrm{Sym}^n(Y) = \{f \colon Y \to \mathbb{N} \cup \{0\} \text{ such that } \sum_{y \in Y} f(y) = n\} \tag{7}$$

with the action of G given by $g.f(y) = f(g^{-1}y)$.

Take $Y = G/H$ for some subgroup $H \subset G$. $\sigma^n(Y) := [\mathrm{Sym}^n(Y)]$ defines a pre-λ structure on $B(G)$:

$$\sigma_t(Y) = \sum_{n=0}^{\infty} \sigma^n(Y) t^n \in 1 + B(G)[[t]].$$

Let $\lambda_t(Y) = \sum\limits_{n=0}^{\infty} \lambda^n(Y) t^n \in 1 + B(G)[[t]]$ be the opposite pre-λ structure, that is

$$\sigma_t(Y)\lambda_{-t}(Y) = \left(\sum_{n=0}^{\infty} \sigma^n(Y) t^n\right)\left(\sum_{n=0}^{\infty} \lambda^n(Y)(-t)^n\right) = 1$$

Is was shown in [BR], that $\lambda^k(Y) = 0$ for $k > |Y|$, so it follows that $\lambda_t(Y)$ is a polynomial of degree at most $|Y|$, hence $\sigma_t(Y)$ is a rational function.

■

Now we are ready to prove the main theorem.

Proof:(of theorem Theorem 2.1)
If $f\colon X \dashrightarrow Y$ is a birational map and both X and Y are smooth and complete, then according to [AKMW, Theorem 0.3.1] $f = \varphi_n \circ \varphi_{n-1} \circ \cdots \circ \varphi_2 \circ \varphi_1$ is a composition of blow-ups and blow-downs φ_i with smooth centres Z_i:

$$X = X_0 \overset{\varphi_1}{\dashrightarrow} X_1 \overset{\varphi_2}{\dashrightarrow} \cdots \overset{\varphi_i}{\dashrightarrow} X_i \overset{\varphi_{i+1}}{\dashrightarrow} \cdots \overset{\varphi_{n-1}}{\dashrightarrow} X_{n-1} \overset{\varphi_n}{\dashrightarrow} X_n = Y$$

so it is enough to prove that such blow-ups and blow-downs preserve rationality of the motivic zeta-function.

In the $K_0(Var_k)$ we have the following relation for $Bl_{Z_i}X_i$ with the exceptional divisor E_i because $Bl_{Z_i}X_i \setminus E \to X_i \setminus Z_i$ is an isomorphism:

$$[Bl_{Z_i}X_i] - [E_i] = [X_i] - [Z_i].$$

Since the motivic zeta-function is a homomorphism we then have:

$$Z_{Bl_{Z_i}X_i}(t) = \frac{Z_{E_i}(t)}{Z_{Z_i}(t)} Z_{X_i}(t) \tag{8}$$

and

$$Z_{X_i}(t) = \frac{Z_{Z_i}(t)}{Z_{E_i}(t)} Z_{Bl_{Z_i}X_i}(t) \tag{9}$$

If follows from (8), (9) and Proposition 1.4 that if both $Z_{E_i}(t)$ and $Z_{Z_i}(t)$ are rational, then these blow-ups and blow-downs preserve rationality of motivic zeta-function (i.e. $Z_{Bl_{Z_i}X_i}(t)$ is rational iff $Z_{X_i}(t)$ is rational).

Rationality of both $Z_{Z_i}(t)$ and $Z_{E_i}(t)$ follows from Lemmas 2.1 and 2.2 respectively because the only non-trivial case is $\dim Z_i = 0$ (blow-up of a smooth variety in a smooth centre of codimension one is an isomorphism), so we are done. ■

3 Motivic zeta-function of Severi-Brauer varities

3.1 Severi-Brauer varities

We only give the definitions and basic facts here. More details one can read: on Severi-Brauer varieties in [GS, Ch.5] and [K3]; on the Grothendieck ring of varieties and motivic zeta-function in [M].

Definition 3.1. A variety P of dimension n over a field k with separable closure k^s is called *Severi-Brauer variety* if $P_{k^s} := P \times_k \operatorname{Spec} k^s \cong \mathbb{P}^n$.
P is called *trivial* if $P \cong \mathbb{P}^n$.
The *index* of a Severi-Brauer variety P is the gcd of degrees all the 0-cycles on P and is denoted by $\operatorname{ind}(P)$.
A subvariety $Y \subset P$ is called *twisted linear subvariety* if $Y_{k^s} \subset \mathbb{P}^n_{k^s}$ is a linear subvariety. Given any subvariety $Y \subset P$ we denote by $\langle Y \rangle$ the minimal twisted linear subvariety containing Y (i.e. $\langle Y \rangle_{k^s}$ is the linear span of Y_{k^s}).

We briefly recall some basic facts:

Proposition 3.1. *Let P be an arbitrary Severi-Brauer variety with* $\operatorname{ind}(P) = i$. *Then*

(1) P is smooth, projective and geometrically irreducible.

(2) P is trivial iff $P(k) \neq \emptyset$.

(3) i divides $(\dim P + 1)$

(4) P has a 0-cycle Z of degree i and $\dim\langle Z\rangle = i - 1$.

(5) Minimal (by inclusion) twisted linear subvaraities are isomorhic to each other and have dimension $i-1$; we denote their isomorphism class by P^{min}.

(6) Severi-Brauer varieties P_1 and P_2 are called let Brauer-equivalent *iff $P_1^{min} \cong P_2^{min}$. We denote this by $P_1 \overset{br}{\sim} P_2$. This holds iff the smaller dimensional one is isomorphic to a twisted linear subvariety of the other*

Remark 1. It is worth noting that if P is a non-trivial Severi-Brauer variety, $\dim P = n$, then $[P] \neq [\mathbb{P}^n]$. It is a corollary of Theorem 1.1, because if $[P] = [\mathbb{P}^n] \in K_0(Var_k)$, then they are also equal in $K_0(Var_k)/(\mathbb{L})$, but $P \times \mathbb{P}^m$ is not birational to $\mathbb{P}^{n+m}$ for any m since $[P]$ does not have a rational point.

The following lemma shows that in dimension one all Severi-Brauer varieties have a nice description:

Lemma 3.1. *If P is a Severi-Brauer curve, then it is isomorphic to a smooth conic C.*

Proof: P is smooth, projective and geometrically irreducible due to Proposition 3.1 so it only remains to compute its genus:

$$g(P) = H^0(P, \Omega_P) = H^0(P_{k^s}, \Omega_{P_{k^s}}) = H^0(\mathbb{P}^1_{k^s}, \Omega_{\mathbb{P}_{k^s}}) = 0$$

and it follows from the Riemann-Roch theorem that $h^0(P, -K_P) = 3$ and $\deg(-K_P) = 2$ so P can be realized as a smooth projective curve in $\mathbb{P}^2$ with $g(P) = \dfrac{(d-1)(d-2)}{2} = 0$ by the genus-degree formula for plane curves so either $\deg(P) = d = 1$ and $P \cong \mathbb{P}^1$ is trivial and isomorphic to a smooth projective conic $V(z^2 - xy)$ containing a rational point $[1:1:1]$ or $d = 2$ and P is isomorphic to a smooth projective conic by the definition. ■

An explicit expression for motivic zeta-function of a smooth conic (i.e. a one-dimensional Severi-Brauer variety according to Lemma 3.1) is well-known:

Proposition 3.2. *Let C be a smooth conic over k with $C(k) = \emptyset$, then $Z_C(t)$ is rational and*

$$Z_C(t) = \frac{1 + [C]t + \mathbb{L}t^2}{(1-t^2)(1-\mathbb{L}^2t^2)}. \tag{10}$$

Proof: It is known that $C_{k^s} \cong \mathbb{P}^1_{k^s}$ hence it is a Severi-Brauer curve. $\mathrm{Sym}^n(C)$ is also a Severi-Brauer variety for any n since base change commutes with symmetric products and $\mathrm{Sym}^n(\mathbb{P}^1) \cong \mathbb{P}^n$.
$\mathrm{Sym}^{2n}(C) \cong \mathbb{P}^{2n}$ because $\deg(-K_C) = 2$ so there is a degree 2 effective zero-cycle on C hence $\mathrm{Sym}^{2n}(C)$ contains a rational point so it is a trivial Severi-Brauer variety.
$\mathrm{Sym}^{2n+1}(C)$ contains C as a twisted linear subvatiety, so they are Brauer-equivalent and it follows from Lemma 3.2 that

$$[\mathrm{Sym}^{2n+1}(C)] = [C](1 + \mathbb{L}^2 + \mathbb{L}^4 + \cdots + \mathbb{L}^{2n}).$$

We have

$$[\mathbb{P}^{2n}] = \sum_{k=0}^{2n} \mathbb{L}^k = \frac{1-\mathbb{L}^{2n+1}}{1-\mathbb{L}}$$

and

$$[C](1+\mathbb{L}^2+\mathbb{L}^4+\cdots+\mathbb{L}^{2n}) = [C]\frac{1-\mathbb{L}^{2n+2}}{1-\mathbb{L}^2}.$$

The motivic zeta-function consists of two parts. Namely, the even part:

$$\sum_{k=0}^{\infty}[\mathbb{P}^{2k}]t^{2k} = \sum_{n=0}^{\infty}\frac{1-\mathbb{L}^{2n+1}}{1-\mathbb{L}}t^{2k} = \frac{1}{1-\mathbb{L}}\left(\frac{1}{1-t^2}-\frac{\mathbb{L}}{1-\mathbb{L}^2t^2}\right) = \frac{1+\mathbb{L}t^2}{(1-t^2)(1-\mathbb{L}^2t^2)}$$

and the odd part:

$$[C]t\sum_{k=0}^{\infty}\frac{1-\mathbb{L}^{2n+2}}{1-\mathbb{L}^2}t^{2n} = \frac{[C]t}{1-\mathbb{L}^2}\left(\frac{1}{1-t^2}-\frac{\mathbb{L}^2}{1-\mathbb{L}^2t^2}\right) = \frac{[C]t}{(1-t^2)(1-\mathbb{L}^2t^2)}$$

summing them up we obtain a rational function

$$Z_C(t) = \frac{1+[C]t+\mathbb{L}t^2}{(1-t^2)(1-\mathbb{L}^2t^2)}. \tag{11}$$

■

The following lemma is a very useful tool for computation of the motivic zeta-function of Severi-Brauer varieties:

Lemma 3.2. *Let X be a minimal Severi-Brauer variety (i.e. does not containing any proper twisted linear subvarieties) of* $\dim X = n$ *and let Y be a Severi-Brauer variety of* $\dim Y = m$ *such that* $X \overset{br}{\sim} Y$. *Then*

$$[Y] = [X]\frac{1-\mathbb{L}^{m+1}}{1-\mathbb{L}^{n+1}} \tag{12}$$

Proof: If $m = n$ there is nothing to prove, so we can assume $m > n$ because X is minimal hence we have $Y^{\min} \cong X$, i.e. X is a minimal twisted linear subvariety in Y. From Proposition 3.1 it follows that $(m+1)$ is divided by $(n+1) = \mathrm{ind}(X)$ and (12) follows directly from [L, Corollary 26]. ■

3.2 Motivic zeta-function

In this part we prove some theorems on motivic-zeta function of a Severi-Brauer variety. The main theorem states that the motivic-zeta function of a Severi-Brauer variety is always rational over $K_0(Var_k)/(\mathbb{L})$. One of the most crucial part of the proof is the stable birational classification of all symmetric powers of a Severi-Brauer variety due to [K1]. However, classes of the symmetric powers of Severi-Brauer varieties in $K_0(Var_k)$ are still understood very poorly, so there is no proof of the rationality of motivic zeta-functions of a Severi-Brauer variety over $K_0(Var_k)$, but we expect that this is true.

Denote by $\langle X\rangle$ the image of the $[X]$ in $K_0(Var_k)/(\mathbb{L})$ and by $\widetilde{Z}_X(t)$ the image of the $Z_X(t)$ in $1+tK_0(Var_k)/(\mathbb{L})[[t]]$.

Theorem 3.1. *Let P be a Severi-Brauer variety over a field of characteristic zero with* $\mathrm{ind}(P) = n$. *Then*

$$\widetilde{Z}_P(t) = \sum_{m=0}^{n-1} \frac{\langle \mathrm{Sym}^{(m,n)}(P)\rangle t^m}{1-t^n} \tag{13}$$

in particularly, it is rational.

Proof: It was proved in [K1, Theorem 1.4] that $\mathrm{Sym}^d(P) \overset{s}{\sim} \mathrm{Sym}^{(d,n)}(P)$ (stably birational) for all $d \geqslant 0$.
Since by Theorem 1.1 the Grothendieck ring $K_0(Var_k)/(\mathbb{L})$ is generated by stable birational equivalence classes of irreducible algebraic varieties we have

$$\langle \mathrm{Sym}^d(P)\rangle = \langle \mathrm{Sym}^{(d,n)}(P)\rangle = \langle \mathrm{Sym}^{(d-n,n)}(P)\rangle = \langle \mathrm{Sym}^{(m,n)}(P)\rangle$$

where $d \equiv m \pmod n$. Then

$$\widetilde{Z}_P(t) = \sum_{m=0}^{n-1}\sum_{k=0}^{\infty}\langle \mathrm{Sym}^{kn+m}(P)\rangle t^{kn+m} = \sum_{m=0}^{n-1}\sum_{k=0}^{\infty}\langle \mathrm{Sym}^{(m,n)}(P)\rangle t^{kn+m} = \sum_{m=0}^{n-1}\frac{\langle \mathrm{Sym}^{(m,n)}(P)\rangle t^m}{1-t^n}$$

■

The following theorem shows that in order to prove rationality of motivic zeta-function of Severi-Brauer varieties over $K_0(Var_k)$, it is enough to consider only the case of minimal Severi-Brauer varieties (i.e. not containing any proper twisted linear subvarieties).

Theorem 3.2. *Fix a natural number n. If motivic zeta-function $Z_X(t)$ is rational for any $(n-1)$-dimensional Severi-Brauer variety X of index n then $Z_Y(t)$ is rational for any Severi-Brauer variety Y of index n.*

Proof: Let Y be a Severi-Brauer variety of $\dim Y = m$ and $\mathrm{ind}(Y) = n$ and $X \subset Y$ be its minimal twisted linear subvariety. Then $\dim X = n-1$ and $\mathrm{ind}(X) = n$ so $Z_X(t)$ is rational by the assumption. It follows from Lemma 3.2 that $(m+1)$ is divided by n.

$$[Y] = [X]\frac{1-\mathbb{L}^{m+1}}{1-\mathbb{L}^n} = [X](1+\mathbb{L}^n+\mathbb{L}^{2n}+\cdots+\mathbb{L}^{\frac{m+1-n}{n}})$$

so using Proposition 1.3 we have the following equality:

$$Z_Y(t) = \prod_{k=0}^{\frac{m+1-n}{n}} Z_X(\mathbb{L}^k t)$$

hence $Z_Y(t)$ is rational whenever $Z_X(t)$ is. ■

The next theorem provides an example of a class of Severi-Brauer varieties with a rational motivic zeta-function.

Theorem 3.3. *Let X be a Severi-Brauer variety with* $\mathrm{ind}(X) = 2$ *and* $\dim X = 2n-1$ *$n \geqslant 1$. Then it contains a unique (up to isomorphism) smooth conic with no rational points C, and $Z_X(t)$ is rational and satisfies the equation*

$$Z_X(t) = \prod_{k=0}^{n-1} Z_C(\mathbb{L}^{2k}t) \tag{14}$$

Proof: If $n = 1$ the claim is equivalent to Lemma 3.1 and Proposition 3.2 so assume that $n \geqslant 2$. By Proposition 3.1 there exists a twisted linear subvariety $P^{min} \subset X$ such that $\dim P = \mathrm{ind}(P) - 1 = 1$ which is a smooth projective conic again by Lemma 3.1. $C(k) = \emptyset$ because otherwise it would not be minimal. Then by Lemma 3.2 we have

$$[X] = [C]\frac{1-\mathbb{L}^{2n}}{1-\mathbb{L}^2} = [C](1+\mathbb{L}^2+\cdots+\mathbb{L}^{2(n-1)})$$

Since $Z_X(t)$ is a homomorphism it follows from (4) that

$$Z_X(t) = \prod_{k=0}^{n-1} Z_C(\mathbb{L}^{2k}t)$$

and rationality follows from the fact that $Z_C(t)$ is rational by Proposition 3.2. ■

Remark 2. It was shown in [S] that over a perfect field the only surfaces that could be birational to a non-trivial Severi-Brauer surface are its dual, a del Pezzo surface of degree 3 and Picard rank 2 and a del Pezzo surface of degree 6 and Picard rank 2 or 3. Together with Theorem 2.1, it tells us that one can try checking rationality of motivic zeta-function on these classes of surfaces instead of Severi-Brauer surfaces. However, the author does not know yet any results on rationality of the motivic zeta-function of such del Pezzo surfaces.

4 Motivic zeta-function of Cartesian power of a conic

This section is devoted to the proof of the following theorem:

Theorem 4.1. *Let C be a smooth conic. Then*

$$Z_{C^{n+1}}(t) = \prod_{k=0}^{n} Z_C^{\binom{n}{k}}(\mathbb{L}^k t) \tag{15}$$

and it is a rational function.

Proof: We prove the theorem by induction on n.
Induction base: We have $[C\times C] = [C\times\mathbb{P}^1]$ according to [K2, Example 6]. Then for $n = 1$ one obtains $Z_{C\times C}(t) = Z_{C\times\mathbb{P}^1}(t)$ because the motivic zeta-function is a homomorphism. It follows from Proposition 1.3, (5) that $Z_{C\times C}(t) = Z_{C\times\mathbb{P}^1}(t) = Z_C(t)Z_C(\mathbb{L}t)$.
Induction step: Assume that

$$Z_{C^n}(t) = \prod_{k=0}^{n-1} Z_C^{\binom{n-1}{k}}(\mathbb{L}^k t).$$

Since $[C^{n+1}] = [C^{n-1}][C\times C] = [C^{n-1}][C\times\mathbb{P}^1] = [C^n\times\mathbb{P}^1]$ we have

$$\begin{aligned} Z_{C^{n+1}}(t) = Z_{C^n}(t)Z_{C^n}(\mathbb{L}t) &= \prod_{k=0}^{n-1} Z_C^{\binom{n-1}{k}}(\mathbb{L}^k t)\prod_{k=0}^{n-1} Z_C^{\binom{n-1}{k}}(\mathbb{L}^{k+1}t) = \\ &= Z_C(t)\prod_{k=1}^{n-1} Z_C^{\binom{n-1}{k}}(\mathbb{L}^k t)\prod_{k=0}^{n-2} Z_C^{\binom{n-1}{k}}(\mathbb{L}^{k+1}t)Z_C(\mathbb{L}^n t) = \\ &= Z_C(t)\prod_{k=1}^{n-1} Z_C^{\binom{n-1}{k}+\binom{n-1}{k-1}}(\mathbb{L}^k t)Z_C(\mathbb{L}^n t) = \prod_{k=0}^{n} Z_C^{\binom{n}{k}}(\mathbb{L}^k t). \end{aligned}$$

Rationality of $Z_{C^{n+1}}(t)$ then follows from Proposition 3.2 ■

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